Nouqta - Calligraphy Engine Fix (animation, dots, ant, editor)

You are a senior frontend engineer. Follow the Nouqta blueprint PDF
(Sections 2, 7, 15.1, 16). All learner text stays in French. This pass is
about the calligraphy engine only. Do NOT touch unrelated screens and do
NOT add features that are not listed here (put ideas in "Suggestions").

RULES

- For every bug: write the ROOT CAUSE in 1-2 sentences first, then fix.
- Do not say "done" unless the numbers in the self-check were actually
  measured and printed.
- Work in the phases below, in order, and report after each.

PHASE A - THE ANIMATION DOES NOT PLAY

What the screenshot shows

On "Voir" the letter is already fully drawn and the ant/start marker sits
at the START of the stroke. So two renderers disagree: the stroke uses
the final state while the ant uses progress = 0. The timeline never
advances.

Likely causes (check ALL of them and report which one it was)

1. **Reduced motion.** An earlier rule said to respect prefers-reduced-motion. If the phone has "Remove animations" on (or
   animation scale = 0), the animation jumps to its final state. FIX:
   the stroke-order animation is lesson CONTENT, not decoration. It must
   play even when reduced motion is on. Use reduced motion only for
   decorative effects (confetti, mascot bounces, page slides). Add a
   Reglages toggle "Animations" (default on) and a small hint in Voir if

the OS reports reduced motion: "Animations reduites : appuie sur
Rejouer".

2. **Two sources of truth.** Replace them with ONE `progress` value
(0..1) per glyph form that drives the stroke dashoffset, the ant
position, the ant rotation and the dot reveal.
3. **Measuring a hidden path.** getTotalLength() returns 0 when the
element is not mounted or display:none. Measure after mount and when
visible.
4. **State per frame.** Do NOT update React state every frame. Run a
requestAnimationFrame loop that writes to refs/DOM styles directly.
5. **Timers/cleanup.** Cancel the rAF on unmount, on step change and on
replay; never run two loops.
6. Step change remounting the SVG, or Rejouer resetting state without
restarting the loop.

Required behaviour

- On entering Voir: show the EMPTY light guide (grey outline of the
letter), then auto-play from progress 0. The

finished letter is shown
only at the end.

- Timeline: strokes one after another, duration proportional to length
(about 700 ms per 100 units, min 500 ms), 250 ms pause between strokes,
then the dots (Phase C).
- Controls with labels: Rejouer, Vitesse (x1 / x0.5, label "Vitesse",
not just "1X"), Pas a pas (one stroke per tap, then dots).
- Start badge "1", "2" appears OFFSET above-left of the start point so
the ant never hides it.

Dev debug overlay (hidden)

Triple-tap the letter title to show: progress, playing state, active
stroke, reducedMotion flag, path lengths, FPS. If Play is pressed and
progress has not moved within 300 ms, show a red message with the likely
reason.

Acceptance (measure and print)

On Baa, Taa, Jiim, Kaaf, Alif: log progress at t = 0, 500 ms, 1500 ms and

end after pressing Rejouer. Values must increase. Repeat 5 times in a
row. Repeat with reducedMotion forced to true: the animation still plays.

PHASE B - VISUAL FIXES IN THE LESSON SCREEN

1. **Stroke too thick.** The letter looks like a fat blob. Use a stroke
   width of about 7-9 percent of the glyph height (about 14-18 in a
   200-unit viewBox), round start, slightly tapered end (reed-pen look).
   The dot size follows from it (Phase C).
2. **Duplicate labels.** "ETAPE 1/5 - OBSERVER", the step bar and the
   title "Observer Jiim" all say the same. Keep ONLY the 5-step bar with
   icons (current step expanded) and the letter name at the top right.
   Remove the other two.
3. **Wrong step states.** Steps Ecouter, Tracer, Ecrire, Retenir show
   green checks while the learner is on step 1. A check may appear only
   for steps really completed. Find why (stale/saved

progress or a bug)
and fix it; report the cause.

4. Keep the description line but max 2 short lines.

PHASE C - DOTS IN THE RIGHT PLACE

The Jiim dot touches the stroke and reads as a blob. Dots must be placed
by rules and verified automatically.

Dot rules

- A dot is a small diamond (square rotated 45 degrees), size about 0.9 x
  stroke width.
- **Clearance (automatic check):** distance from the dot center to the
  stroke centerline >= strokeWidth/2 + dotHalfDiagonal + 6 units. Dots
  must not overlap each other (min gap 6 units) and must stay inside the
  viewBox with 12 units of padding.
- Placement per letter (isolated form):
  - Baa: 1 dot below, centered under the belly.
  - Taa: 2 dots above, side by side, centered.

    - Thaa: 3 dots above, triangle with the apex UP (one over two).
    - Jiim: 1 dot INSIDE the lower curve (the belly cavity), centered in the
      cavity, in the lower half of the bowl, clearly separated from the
      stroke by at least one dot diameter. It must NOT sit on the baseline
      crossing or touch the stroke.
    - Haa: none. Khaa: 1 dot above the hook, centered over the body.
    - Dhaal, Zaay, Daad, Dhaa-fort, Ghayn, Faa, Nuun: 1 dot above, centered
      over the main body (Faa: over the loop; Nuun: over the bowl).
    - Shiin: 3 dots above, triangle apex UP.
    - Qaaf: 2 dots above, side by side, over the loop.
    - Yaa: 2 dots below, side by side.
- Store dots as data (position, arrangement, size) per glyph form. Fix
  the Jiim data now.

Dot animation

After the last stroke, the ant carries the dot (a crumb) from the side,
walks to its place and drops it with a small pop (scale

1.3 -> 1) and a
soft haptic. Multiple dots: one by one.

Acceptance

The clearance check runs for all 28 letters in
/dev/selftest and prints
PASS/FAIL with the distance for each dot.

PHASE D - THE ANT MOVES (identity of the app)

Build the ant as a small SVG (about 40 px) with real motion:

- **Walk cycle:** 6 legs in two alternating groups, switching every
  120 ms (CSS steps animation) while it moves; antennae wiggle.
- **Follows the stroke:** position from getPointAtLength(progress), body
  rotated to the tangent (angle from two nearby samples). Move with
  transform translate/rotate on a ref (no per-frame React state).
- **Behaviour:** short pause (150 ms) at corners and stroke ends, small
  bounce when a stroke finishes, happy spin/jump

when the whole letter is
done (Kids), carries dots (Phase C).

- **Ink trail:** the drawn stroke is the trail; it comes from behind the
ant.
- **Home:** the ant idles on the current node (looks around, wiggles),
walks to the next node when a letter is finished, wiggles when tapped.
- Adults: smaller, no jumping; the ant is still the pen tip.
- Reduced motion: decorative ant motion (idle, jumps) becomes static; the
walking along the stroke in Voir still works (Phase A rule).

PHASE E - STROKE EDITOR (so the owner can fix letters without code)

Hidden route /dev/glyphs (also opened by tapping the logo 5 times).

- Letter list with a status per letter: Brouillon / Valide.
- Canvas showing the stroke path with draggable anchors and bezier
handles; add/remove points; undo/redo.

- **Reference overlay:** the real letter in the Amiri font at 25 percent
  opacity behind the strokes, toggle on/off, to align the shapes (this is
  for authoring only, never for tracing).
- Stroke order list (drag to reorder), direction arrow, start point.
- Dot editor: drag dots, choose count and arrangement (row, triangle up).
- Validation panel: dot clearance, strokes inside the viewBox, number of
  steps (bodyStrokes + dots step). Red/green.
- Button "Jouer l'animation" runs the REAL animation engine.
- Saves to local storage; "Copier le JSON" / "Importer un JSON" for one
  letter or all. The app loads Valide overrides first.
- Mark every seeded glyph as Brouillon until validated.

PHASE F - GAMES CHANGE EVERY TIME, MORE INTERACTIVE

1. **Generators, not fixed lists.** Build questions and rounds at runtime
   from letter data + templates with seeded

randomness (seed = date +
session counter). No item repeated within the last 20. Answer
positions shuffled; the correct answer never in the same position 3
times in a row. Question types and games never repeat twice in a row.
Weight 60/40 toward letters the learner got wrong most.

2. **Offline.** Generation is client-side from bundled data, no network
needed. If the environment allows, add a PWA service worker (cache app
shell, data, fonts, ant sprites), an "Hors ligne" indicator, and a
queue for photo/voice uploads. If it is not possible here, say so in
the report.
3. **More interactive (gestures, not just taps):**
    - Memoire: flip with animation, matched pairs fly to the side.
    - Course de la fourmi: tap the stone, the ant hops with a real arc.
    - Chasse aux points: DRAG dots onto the letter, snapping to the
      correct places.
    - Mot a assembler: drag tiles with snapping and a bounce on success.

    - Lettre mystere: swipe/hold to reveal strokes, fewer reveals = more
      stars.
      Every game has sound + haptics, a combo meter (flame), ant reactions,
      and 3 variants of French feedback sentences so they do not repeat.

PHASE G - HARAKAT LESSON (minimal)

- Keep the short vowels (a/i/u) in step Ecouter for every letter.
- After Family 1 is finished: a dedicated lesson "Les voyelles courtes"
  with 3 cards (fatha above, kasra below, damma with a loop), the ant
  drawing each mark, syllable reading of the 4 letters of Family 1 with
  each mark, then the quiz type "Lecture voyelle".
- After all 28 letters: locked entry "Voyelles et signes" (sukun, shadda,
  tanwin, long vowels), labelled "Bientot". Do not build it now.
- Use combining characters U+064E, U+0650, U+064F; line-height at least
  1.9 so marks never clip.

MANDATORY SELF-CHECK BEFORE YOU SAY "DONE"

Update /dev/selftest to print a PASS/FAIL table, with the measured
numbers:

1. Animation progress at 0 / 500 / 1500 ms / end for Baa, Taa, Jiim,
   Kaaf, Alif; 5 replays each; also with reducedMotion forced on.
2. Dot clearance for all 28 letters (distance values printed).
3. Dot count/position matches the table; Jiim dot is inside the cavity.
4. Step bar shows checks only for completed steps (test a brand new
   user).
5. No duplicate step labels; stroke width within 7-9 percent of glyph
   height.
6. Ant: rotates with the tangent (log the angle at 3 points), walk cycle
   runs only while moving.
7. Editor: change a Jiim dot, save, reload, the change persists and the
   animation uses it.

8. Games: 20 consecutive generated questions have no repeats and varied
   types/positions; works with the network disabled.
9. Tap targets >= 48px, no overlap at 360x740 and 390x844, no English
   words in the UI.
   End your reply with: (a) the table, (b) every failure and its fix, (c)
   the ROOT CAUSE of the animation bug, (d) anything you could NOT verify,
   stated honestly, (e) a short "Suggestions" list.